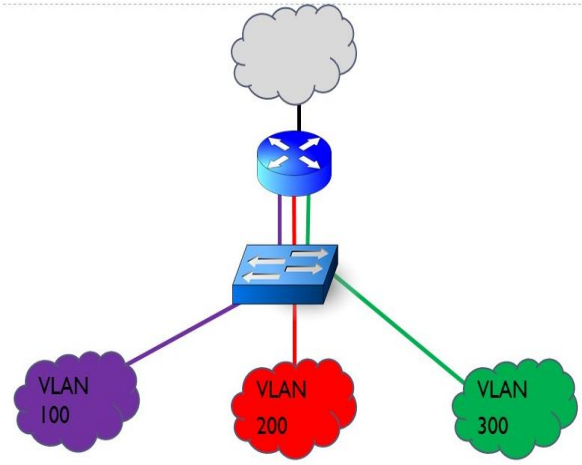


CLOUD COMPUTING COURSE

Lecture16-Network Virtualization in CC (5)

ABSTRACT

Lecture16 Introduces Virtual Networks -VLANs

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Network Virtualization

Definition of NW Virtualization

Network virtualization (NV) is defined by the ability to create logical, virtual networks that are decoupled from the underlying network hardware to ensure the network can better integrate with and support increasingly virtual environments.

Network virtualization abstracts networking connectivity and services that have traditionally been delivered via hardware into a logical virtual network that is decoupled from and runs independently on top of a physical network in a hypervisor.

Applying Virtualization to the Network

When applied to a network, virtualization creates a logical software-based view of the hardware and software networking resources, such as switches and routers. The physical networking devices are simply responsible for the forwarding of packets, while the virtual network (software) provides an intelligent abstraction that makes it easy to deploy and manage network services and underlying network resources. As a result, it can align the network to better support virtualized environments.

How NW Virtualization Solve Network Challenges?

Network performance is an important factor in the productivity of an organization. One of the technologies used to improve network performance is the separation **of large broadcast domains** into smaller ones. By design, routers will block broadcast traffic at an interface. However, routers normally have a limited number of LAN interfaces. A router's primary role is to move information between networks, not provide network access to end devices. The role of providing access into a LAN is normally reserved for an access layer switch.

A **virtual local area network (VLAN**) can be created on a Layer 2 switch to reduce the size of broadcast domains, similar to a Layer 3 device.

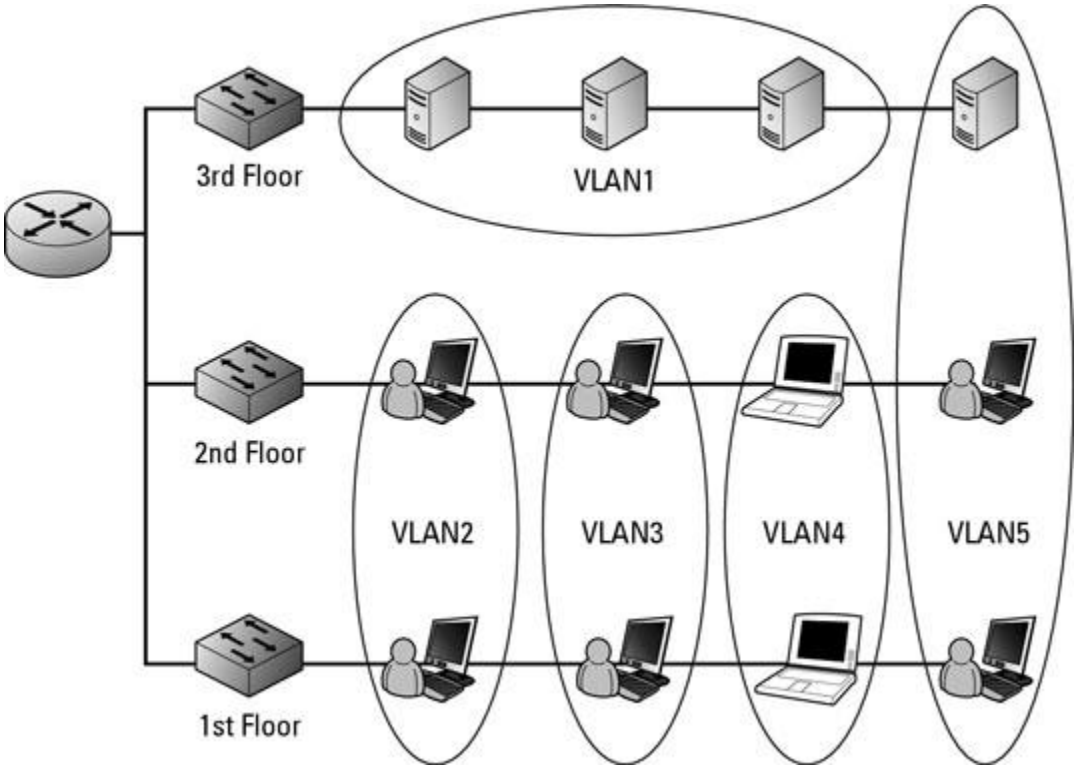
VLANs are commonly incorporated into network design making it easier for a network to support the goals of an organization. While VLANs are primarily used within switched local area networks, modern implementations of VLANs allow them to span MANs and WANs.

Virtual local area network (VLAN)

It is a logical subgroup within a local area network that is created via software rather than manually moving cables in the wiring closet. It combines user stations and network devices into a single unit regardless of the physical LAN segment they are attached to and allows traffic to flow more efficiently within populations of mutual interest.

- VLANs are implemented in LAN **switches (Layer 2 device).**
- VLANs reduce the time it takes to implement, moves, adds and changes.

- VLANs allow an administrator to segment networks based on factors such as function, project team, or application, **without regard for the physical** location of the user or device.



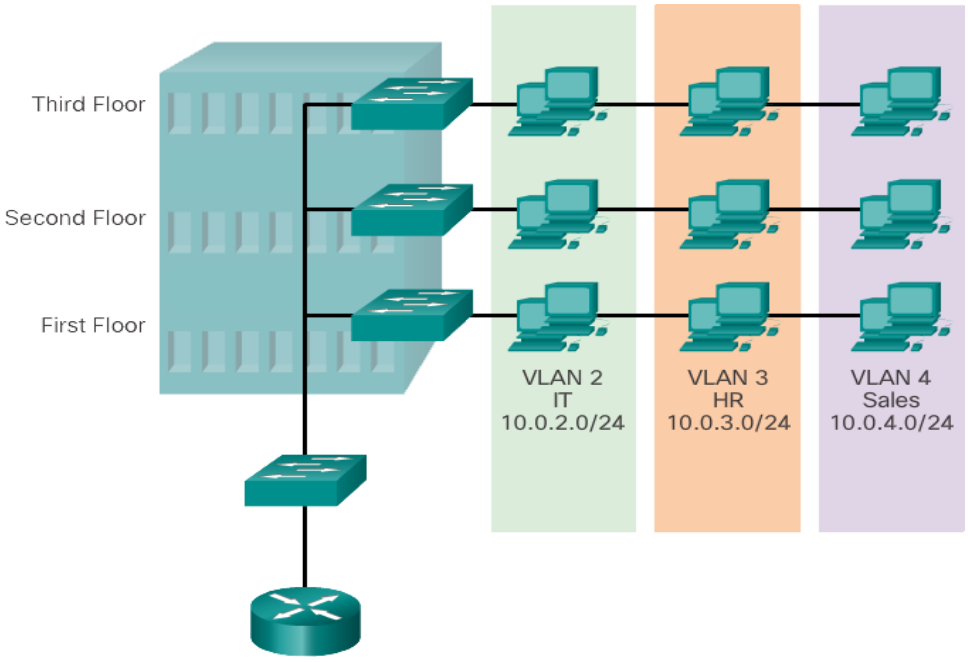
- Devices within a VLAN act as if they are in their own independent network, even if they share a common infrastructure with other VLANs.

- Any **switch port can belong to a VLAN**, and unicast, broadcast, and multicast packets are forwarded and flooded only to end stations within the VLAN where the packets are sourced.

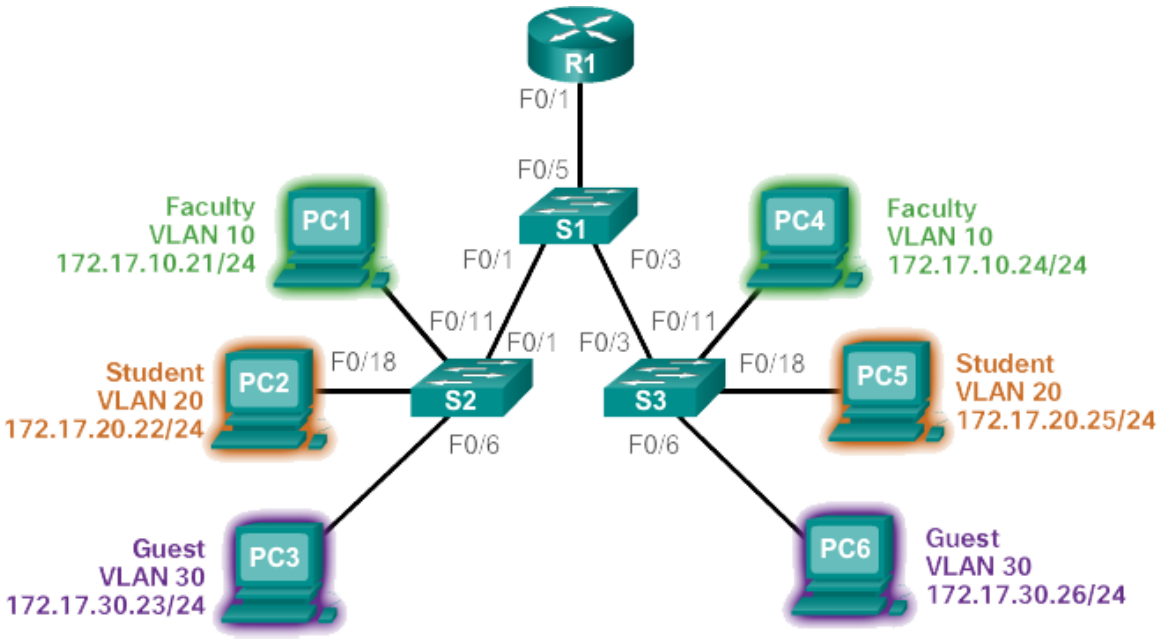
- Each **VLAN is considered a separate logical network**, and packets destined for stations that do not belong to the VLAN must be forwarded through a device that supports routing.

- A VLAN creates a logical broadcast domain that can span multiple physical LAN segments. VLANs improve network performance **by separating large broadcast domains into smaller ones. If a device in one VLAN sends a broadcast Ethernet frame, all devices in the VLAN receive the frame, but devices in other VLANs do not.**

- VLANs enable the implementation of access and security policies according to specific groupings of users. **Each switch port can be assigned to only one VLAN**



- Each VLAN in a switched network corresponds to an IP network, which means that IP network numbers are applied to VLANs in an orderly fashion that takes the network as a whole into consideration.



VLAN Types

Most VLANs will be one of five main types, depending on their purpose:

1. **Management VLAN:** A best practice is to set up a separate VLAN for **management traffic** like monitoring, system logging, and other potentially sensitive management tasks. In addition to the security benefits, this ensures that bandwidth for management will be available even when user traffic is high.

2. **Data VLAN:** Also known as a user VLAN, the data VLAN is designated only for user-generated data. How you group your data VLANs (such as by department or workgroup, for example) will depend on your organization's

structure and business processes. Before you jump into configuring your data VLANs, look at your entire possible VLAN landscape and spend some time assessing the logic for how to best group your users.

3. **Voice VLAN:** If your organization uses voice over IP (VoIP), you will want to have a separate voice VLAN. This will preserve bandwidth for other applications and ensure VoIP quality.

4. **Default VLAN:** Typically, the default VLAN refers to the one that all of the ports on a device belong to when it is switched on. On most switches, this default is VLAN 1 and should be changed for security reasons. Some network managers may use the term “default VLAN” to refer to a VLAN to which all ports are assigned when they are not being used.

5. **Native VLAN:** The native VLAN is the one into which untagged traffic will be put when it is received on a trunk port. This makes it possible for your VLAN to support legacy devices or devices that don’t tag their traffic like some wireless access points and simply network attached devices.

VLAN Connections

During the configuration of VLAN on port, we need to know what type of connection it has. Switch supports **two** types of VLAN connection

- **Access link:** A switch port in access modes belongs to one specific VLAN and sends and receives regular Ethernet frames in untagged form. The switch interfaces connected to devices such as desktops, laptops, printers etc. are typically configured as access ports.

- **Trunk link:** Usually trunk link connection is used to connect two switches or switch to router, it carry traffic from multiple VLANs at the same time.

  VLAN can span anywhere in network, that is happen due **to trunk link** connection. Trunking allows us to send or receive VLAN information across the network. To support trunking, original Ethernet frame **is modified** to carry VLAN information.

